# Lecture 6. Generalized Linear Models and Model Evaluation

Logistic and Poisson regression, prediction error, cross-validation

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## 6.1 Generalized linear models

**Motivation.** GLMs link a linear predictor to the mean of a binary, count or other restricted outcome.

A monotone **link function** $f$:

$$f(\theta_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}, \qquad y_i \sim D(\theta_i, \psi),$$

where $\theta_i$ is the centre of the distribution $D$ and $\psi$ controls its scale or shape.

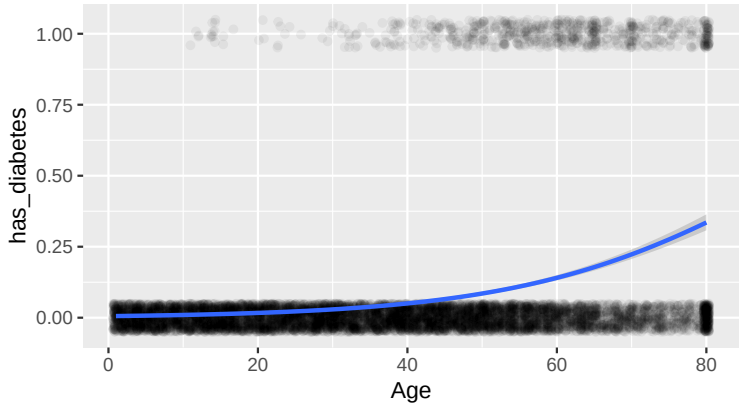
## 6.2 Logistic regression

For binary $y \sim \text{Bernoulli}(\theta)$ the link is the **logit**:

$$\text{logit}(\theta) = \log \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad \theta = \frac{1}{1 + \exp\{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)\}} \in (0, 1).$$

Diabetes against age and BMI (`NHANES`):

```
NHANES <- NHANES %>% mutate(has_diabetes = as.numeric(Diabetes == "Yes"))
NHANES %>% filter(!is.na(has_diabetes)) %>%
  ggplot(aes(Age, has_diabetes)) +
  geom_jitter(height = 0.05, alpha = 0.05) +
  geom_smooth(method = "glm", method.args = list(family = "binomial"), formula = y ~ x)
```




The probability rises with age (and BMI). Both together:

```
logreg <- glm(has_diabetes ~ BMI + Age, family = "binomial", data = NHANES)
tidy(logreg)
```

```
## # A tibble: 3 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)  -8.08     0.244       -33.1 1.30e-239
## 2 BMI           0.0943   0.00552      17.1 1.74e- 65
## 3 Age           0.0573   0.00249      23.0 2.28e-117
```

Predicted probability at age 60, BMI 25:

```
linear_component <- c(1, 25, 60) %*% coef(logreg)
exp(linear_component) / (1 + exp(linear_component))
```

```
##           [,1]
## [1,] 0.09233095
```

```
predict(logreg, newdata = data.frame(BMI = 25, Age = 60), type = "response")
```

```
##         1
## 0.09233095
```

### 6.2.1 Coefficients as odds ratios

A unit change in $x$ does not change the probability by a constant amount, so coefficients are read through **odds** $p/(1-p)$ and **odds ratios**. With a single 0/1 predictor, logit $P(y_i = 1) = \beta_0 + \beta_1 x_i$, the log odds are $\beta_0 + \beta_1$ when $x_i = 1$ and $\beta_0$ when $x_i = 0$, so with $p = P(y = 1 \mid x = 1)$ and $q = P(y = 1 \mid x = 0)$,

$$e^{\beta_1} = \frac{p/(1-p)}{q/(1-q)}.$$

In general $e^{\beta_j}$ multiplies the odds per unit of $x_j$, others fixed:

```
exp(coef(logreg))
```

```
##  (Intercept)          BMI          Age
## 0.0003095805 1.0989173575 1.0589497173
```

Odds $\times 1.06$ per year of age, $\times 1.10$ per BMI unit.

## 6.3 GLMs for counts

**Motivation.** Count variance grows with the mean: Poisson, then negative binomial and zero-inflated models.

### 6.3.1 Poisson regression

A Poisson variable with rate $\lambda$ has $P(x = k \mid \lambda) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$, mean = variance $= \lambda$. Poisson regression: $y_i \sim \text{Poisson}(\lambda_i)$ with the log link,

$$\log \lambda_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip},$$

no $\psi$; $e^{\beta_j}$ multiplies the rate: $\lambda(x_j + 1) = \lambda(x_j)\, e^{\beta_j}$.

**Example.** `pscl::bioChemists`: articles (`art`) of 915 biochemistry PhD students in the last three PhD years, with gender, marriage, kids under 5, prestige, mentor's articles.

```
bioChemists <- bioChemists %>% mutate(gender = ifelse(fem == "Men", 1, 0))
pois_mod <- glm(art ~ gender, family = "poisson", data = bioChemists)
round(coef(pois_mod), 3)
```

```
## (Intercept)      gender
##       0.385       0.247
```

```
exp(coef(pois_mod))
```

```
## (Intercept)      gender
##    1.470309    1.280405
```

Women: $\hat{\lambda} = e^{0.385} = 1.47$; men: $e^{0.385+0.247} = 1.88$, ratio $e^{0.247} = 1.28$. (The slides write $\lambda_{\text{women}} = e^{0.247}$; the women's rate is $e^{\hat{\beta}_0} = e^{0.385}$, which is the value 1.470 given there [Supplement].)

### 6.3.2 Overdispersion and the negative binomial

**Overdispersion**: variance above the model's (for counts, above the mean); a Poisson fit then *underestimates* standard errors.

```
var(bioChemists$art) / mean(bioChemists$art)
```

```
## [1] 2.191358
```

Variance over twice the mean. The **negative binomial** counts failures before the $r$-th success:

$$P(x = k \mid r, \theta) = \binom{r+k-1}{k} \theta^r (1-\theta)^k = \frac{\Gamma(r+k)}{\Gamma(r)\, k!} \theta^r (1-\theta)^k,$$

(Gamma form: non-integer $r$.) `dnbinom(2, 3, 0.75)` $= P(x = 2 \mid r = 3, \theta = 0.75)$. Mean $\mu = r(1-\theta)/\theta$, so $\theta = r/(r+\mu)$; variance $\mu + \mu^2/r$.

```
dnbinom(2, 3, 0.75)
```

```
## [1] 0.1582031
```

It is a Poisson mixture with gamma-distributed means.

**Poisson–gamma mixture [Supplement]**

If $\lambda \sim \text{Gamma}(\text{shape } r,\ \text{rate } r/\mu)$ and $x \mid \lambda \sim \text{Poisson}(\lambda)$, then

$$P(x = k) = \int_0^\infty \frac{e^{-\lambda}\lambda^k}{k!} \frac{(r/\mu)^r \lambda^{r-1} e^{-r\lambda/\mu}}{\Gamma(r)}\, d\lambda = \frac{\Gamma(r+k)}{\Gamma(r)\, k!} \Big(\frac{r}{r+\mu}\Big)^r \Big(\frac{\mu}{r+\mu}\Big)^k,$$

the negative binomial with $\theta = r/(r+\mu)$; $\mu^2/r$ is the variance of the rates.

**Negative binomial regression** assumes $y_i \sim \text{NegBin}(\mu_i, r)$ with $\log \mu_i = \beta_0 + \beta x_i$; `MASS::glm.nb()` fits it:

```
nb_mod <- glm.nb(art ~ gender, data = bioChemists)
tidy(nb_mod)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    0.385    0.0544      7.08 1.44e-12
## 2 gender         0.247    0.0720      3.43 6.01e- 4
```

```
tidy(pois_mod)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    0.385    0.0402      9.59 8.77e-22
## 2 gender         0.247    0.0519      4.77 1.89e- 6
```

Same coefficients, larger SEs: Poisson was overconfident.

### 6.3.3 Zero-inflated Poisson

Excess zeros (cigarettes smoked: non-smokers report 0). **Zero-inflated Poisson** (ZIP): with probability $\theta_i$ a "structural zero" ($z_i = 1$), otherwise Poisson,

$$y_i \sim \theta_i \mathbb{1}(y_i = 0) + (1-\theta_i)\,\text{Poisson}(\lambda_i), \qquad \log \lambda_i = \beta_0 + \beta_1 x_i, \quad \text{logit}\, \theta_i = \gamma_0 + \gamma_1 x_i,$$

Slides (`zeroinfl(cigs ~ educ)`): more education, more non-smokers ($\hat\theta$ 0.49 at 6 years to 0.72 at 18) but heavier smokers ($\hat\lambda$ 18.3 to 27.7). With `bioChemists`:

```
zip_mod <- zeroinfl(art ~ gender | gender, data = bioChemists)
zip_mod
```

```
##
## Call:
## zeroinfl(formula = art ~ gender | gender, data = bioChemists)
##
## Count model coefficients (poisson with log link):
## (Intercept)       gender
##      0.6151       0.2393
##
## Zero-inflation model coefficients (binomial with logit link):
## (Intercept)       gender
##    -1.35416     -0.03902
```

Count part: men publish more when they publish; zero part: gender effect on being a "non-publisher".

## 6.4 Evaluating models

**Motivation.** Training fit is optimistic; honest evaluation needs unseen data.

We seek $g$ with $y = g(x) + \epsilon$ (linear: $g(x) = \beta_0 + \cdots + \beta_p x_p$; GLM: $g(x) = f^{-1}(\beta_0 + \cdots + \beta_p x_p)$).

**A coin example.** Estimating a fair coin's $p$: a uniform random guess has zero bias, large variance; always 0.7 has zero variance, large bias.

**Proposition 6.1 (bias–variance decomposition)**

For a new point $(x, y)$ with $y = g(x) + \epsilon$, $\mathbb{E}\epsilon = 0$, and $\epsilon$ independent of the fitted model $\hat{y}(\cdot)$,

$$\mathbb{E}\big[(y - \hat{y}(x))^2\big] = \mathbb{E}[\epsilon^2] + \big(g(x) - \mathbb{E}[\hat{y}(x)]\big)^2 + \mathbb{E}\big[(\mathbb{E}[\hat{y}(x)] - \hat{y}(x))^2\big].$$

**Proof [Supplement].** $y - \hat{y} = \epsilon + \{g(x) - \mathbb{E}\hat{y}\} + \{\mathbb{E}\hat{y} - \hat{y}\}$; square and take expectations. The middle term is constant; $\mathbb{E}\epsilon = 0$ and independence kill the cross terms with $\epsilon$; $\mathbb{E}\{\mathbb{E}\hat{y} - \hat{y}\} = 0$ kills the last one. ■

**Model risk = observation variance + (model bias)$^2$ + model variance.**

- *Observation variance* $\sigma^2$: irreducible noise.
- *Model bias*: too limited a family (quadratic for cubic data), **underfitting**; reduced by complexity.
- *Model variance*: fit changes a lot across samples, **overfitting**; reduced by simplicity.

With complexity, bias falls and variance rises: risk is U-shaped.

### 6.4.1 Training error and the holdout method

Training error favours complexity. **Holdout**: set data aside. `RailTrail`: hold out every fifth day (18 of 90), fit degrees 1–5 on 72:

```
index <- (1:18) * 5
test <- RailTrail[index, ]
train <- RailTrail[-index, ]
mods <- map(1:5, ~ glm(volume ~ poly(hightemp, .x), data = train))
tibble(degree = 1:5,
       train_mse = map_dbl(mods, ~ mean((train$volume - predict(.x))^2)),
       valid_mse = map_dbl(mods, ~ mean((test$volume - predict(.x, newdata = test))^2)),
       AIC = map_dbl(mods, AIC))
```

```
## # A tibble: 5 x 4
##   degree train_mse valid_mse   AIC
##    <int>     <dbl>     <dbl> <dbl>
## 1      1    11429.     7835.  883.
## 2      2    10277.     6531.  877.
## 3      3     9318.     4975.  872.
## 4      4     9210.     5128.  874.
## 5      5     9203.     5023.  875.
```

Training error keeps falling; validation error and AIC both pick degree 3. The held-out part is a *validation* set; splits 80/20, 90/10, 75/25 are common.

### 6.4.2 Cross-validation

A single split wastes data. **$k$-fold cross-validation**: random $k$ folds, each validating a fit on the other $k-1$, giving $\mathrm{MSE}_1, \ldots, \mathrm{MSE}_k$, and

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i.$$

**Leave-one-out** CV ($k = n$): no randomness, less bias than holdout, $n$ fits. `boot::cv.glm()`:

```
set.seed(1)
cv_error <- map_dbl(1:5, function(i) {
  fit <- glm(volume ~ poly(hightemp, i), data = RailTrail)
  boot::cv.glm(RailTrail, fit, K = 5)$delta[1]
})
cv_error
```

```
## [1] 11386.289  9845.353  8797.032  9319.328  9336.853
```

**LOOCV without refitting [Supplement]**

For least squares, leaving out observation $i$ changes its residual to $e_i/(1-h_{ii})$, with $h_{ii}$ the leverage, so $\mathrm{CV}_{(n)} = \frac{1}{n}\sum_i \{e_i/(1-h_{ii})\}^2$ needs a single fit.

```
fit <- lm(volume ~ poly(hightemp, 3), data = RailTrail)
mean((residuals(fit) / (1 - hatvalues(fit)))^2)
```

```
## [1] 8992.758
```

```
boot::cv.glm(RailTrail, glm(volume ~ poly(hightemp, 3), data = RailTrail))$delta[1]
```

```
## [1] 8992.758
```

### 6.4.3 Why a separate test set

The best of many validation losses is optimistic (a tournament winner was also lucky). Report a **test set** never used for fitting or selection: train to fit, validate (or CV) to select, test once.

## 6.5 Selected exercises

**Motivation.** Three results behind this lecture's numbers.

**Exercise 6.1 (one binary predictor)**

Let $x_i \in \{0,1\}$, with $n_g$ observations and mean response $\bar{y}_g$ in group $g = 0, 1$.
(a) In the logistic regression logit $P(y_i = 1) = \beta_0 + \beta_1 x_i$, suppose $0 < \bar{y}_0, \bar{y}_1 < 1$. Show that the maximum likelihood fit gives $\hat{P}(y = 1 \mid x = g) = \bar{y}_g$, so $e^{\hat{\beta}_1}$ is the sample odds ratio.
(b) In the Poisson regression $\log \lambda_i = \beta_0 + \beta_1 x_i$, suppose $\bar{y}_0, \bar{y}_1 > 0$. Show that $\hat{\lambda}_g = \bar{y}_g$ and $e^{\hat{\beta}_1} = \bar{y}_1/\bar{y}_0$.

**Solution [Supplement].**

(a) Put $p_0 = \operatorname{expit}(\beta_0)$ and $p_1 = \operatorname{expit}(\beta_0 + \beta_1)$; this is a bijection from $\mathbb{R}^2$ onto $(0,1)^2$. The log-likelihood splits into two parts with separate parameters,

$$\ell = \sum_{g=0,1} n_g \{\bar{y}_g \log p_g + (1 - \bar{y}_g) \log(1 - p_g)\}.$$

Each part is strictly concave in $p_g$ with derivative $n_g\{\bar{y}_g/p_g - (1 - \bar{y}_g)/(1 - p_g)\}$, which vanishes only at $p_g = \bar{y}_g \in (0,1)$. So $\hat{p}_g = \bar{y}_g$, and

$$e^{\hat{\beta}_1} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1}}{e^{\hat{\beta}_0}} = \frac{\hat{p}_1/(1 - \hat{p}_1)}{\hat{p}_0/(1 - \hat{p}_0)} = \frac{\bar{y}_1/(1 - \bar{y}_1)}{\bar{y}_0/(1 - \bar{y}_0)}.$$

If some $\bar{y}_g$ is 0 or 1, the supremum is approached only as $p_g \to \bar{y}_g$, i.e. $\hat{\beta} \to \pm\infty$: this is the separation problem of Lecture 9.

(b) With $\lambda_0 = e^{\beta_0}$ and $\lambda_1 = e^{\beta_0 + \beta_1}$ (a bijection onto $(0,\infty)^2$), the log-likelihood is, up to a constant, $\sum_g n_g(\bar{y}_g \log \lambda_g - \lambda_g)$. Each term is strictly concave with derivative $n_g(\bar{y}_g/\lambda_g - 1)$, zero at $\lambda_g = \bar{y}_g$. Hence $\hat{\lambda}_g = \bar{y}_g$ and $e^{\hat{\beta}_1} = \hat{\lambda}_1/\hat{\lambda}_0 = \bar{y}_1/\bar{y}_0$. For `bioChemists` this is why $e^{0.385} = 1.47$ and $e^{0.385+0.247} = 1.88$ are the mean article counts of women and men. ■

**Exercise 6.2 (moments of the zero-inflated Poisson)**

Let $Y = (1 - Z)W$ with $Z \sim \text{Bernoulli}(\theta)$ and $W \sim \text{Poisson}(\lambda)$ independent. Show that

$$P(Y = 0) = \theta + (1 - \theta)e^{-\lambda}, \qquad \mathbb{E}Y = (1 - \theta)\lambda, \qquad \operatorname{Var} Y = (1 - \theta)\lambda(1 + \theta\lambda),$$

and conclude that a ZIP model is overdispersed whenever $\theta > 0$. Why does this mean that overdispersion alone cannot decide between a negative binomial and a ZIP model?

**Solution [Supplement].**

*Step 1.* $Y = 0$ if $Z = 1$, or if $Z = 0$ and $W = 0$; these events are disjoint, so $P(Y = 0) = \theta + (1 - \theta)e^{-\lambda}$.

*Step 2.* Since $(1 - Z)^2 = 1 - Z$ and $Z$, $W$ are independent, $\mathbb{E}Y = (1 - \theta)\mathbb{E}W = (1 - \theta)\lambda$ and $\mathbb{E}Y^2 = (1 - \theta)\mathbb{E}W^2 = (1 - \theta)(\lambda + \lambda^2)$.

*Step 3.* Therefore

$$\begin{aligned}\operatorname{Var} Y &= (1 - \theta)(\lambda + \lambda^2) - (1 - \theta)^2\lambda^2 = (1 - \theta)\lambda\{1 + \lambda - (1 - \theta)\lambda\} \\ &= (1 - \theta)\lambda(1 + \theta\lambda).\end{aligned}$$

*Step 4.* $\operatorname{Var} Y/\mathbb{E}Y = 1 + \theta\lambda > 1$ for $\theta > 0$. The negative binomial ($\operatorname{Var}/\mathbb{E} = 1 + \mu/r$) is also overdispersed, so overdispersion only rejects Poisson. The ZIP puts the extra variability in zeros: compare the observed zero share with $\theta + (1 - \theta)e^{-\lambda}$ and the negative binomial's $P(Y = 0)$. ■

**Exercise 6.3 (LOOCV from a single fit)**

Let $X$ be the $n \times p$ design matrix of a least-squares fit, $A = X^{\mathsf{T}}X$ invertible, $H = XA^{-1}X^{\mathsf{T}}$ with diagonal $h_{ii}$, and residuals $e_i = y_i - \hat{y}_i$. Assume that deleting row $i$ leaves a full-rank design. Prove that the fit without observation $i$ predicts $y_i$ with error

$$y_i - x_i^{\mathsf{T}}\hat{\beta}_{(-i)} = \frac{e_i}{1 - h_{ii}},$$

which justifies the formula for $\mathrm{CV}_{(n)}$ in the note above.

**Solution [Supplement].** Write $x = x_i$, $h = h_{ii} = x^{\mathsf{T}}A^{-1}x$ and $b = X^{\mathsf{T}}y$, so $\hat{\beta} = A^{-1}b$ and $\hat{y}_i = x^{\mathsf{T}}A^{-1}b$.

*Step 1.* ($h < 1$.) $H$ is symmetric and idempotent, so $h = \sum_j h_{ij}^2 \geq h^2$ and $0 \leq h \leq 1$. Deleting row $i$ gives $X_{(-i)}^{\mathsf{T}}X_{(-i)} = A - xx^{\mathsf{T}}$, invertible by assumption; since $\det(A - xx^{\mathsf{T}}) = \det(A)(1 - x^{\mathsf{T}}A^{-1}x)$, we get $h \neq 1$, hence $h < 1$.

*Step 2.* (Sherman–Morrison.) Multiplying out, using $x^{\mathsf{T}}A^{-1}x = h$,

$$(A - xx^{\mathsf{T}})\Big(A^{-1} + \frac{A^{-1}xx^{\mathsf{T}}A^{-1}}{1 - h}\Big) = I + xx^{\mathsf{T}}A^{-1}\Big(\frac{1}{1-h} - 1 - \frac{h}{1-h}\Big) = I,$$

so the second factor is $(A - xx^{\mathsf{T}})^{-1}$.

*Step 3.* (Leave-one-out prediction.) $\hat{\beta}_{(-i)} = (A - xx^{\mathsf{T}})^{-1}(b - xy_i)$. Since $x^{\mathsf{T}}A^{-1}(b - xy_i) = \hat{y}_i - hy_i$,

$$x^{\mathsf{T}}\hat{\beta}_{(-i)} = (\hat{y}_i - hy_i) + \frac{h(\hat{y}_i - hy_i)}{1 - h} = \frac{\hat{y}_i - hy_i}{1 - h}.$$

*Step 4.* Hence $y_i - x^{\mathsf{T}}\hat{\beta}_{(-i)} = \{(1-h)y_i - \hat{y}_i + hy_i\}/(1-h) = e_i/(1-h)$. ∎